

Objective Disability Classification and Judgment

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Science seems most objective when it appears free of human judgment. In this view, scientists should set aside their values, perspectives, and interpretations, allowing evidence alone to determine conclusions. This ideal is appealing because it offers neutrality: if scientists let the evidence lead, scientific claims are less likely to be distorted by personal bias or social pressure. However, this image is hard to sustain when we ask how evidence supports scientific claims. Evidence does not automatically yield data, nor does it determine its own value. Experts must decide what counts as relevant evidence, how reliable it is, and whether it is enough for a claim.

Disability classification is a specific example. Unlike most scientific classifications, which rely on a single type of measurement, disability classification draws on biomedical, functional, social, and historical evidence. The World Health Organization’s International Classification of Functioning, Disability and Health (ICF) states that disability results from the interaction between impairments, activity limitations, participation restrictions, and surrounding factors (World Health Organization 3). Disability is not merely a medical condition limited to an individual’s body. Because disability classifications combine different kinds of evidence and affect people’s lives in major ways, scientists must make judgments about how to interpret and weigh that evidence. These classifications also affect access to accommodation, services, protections, and recognition. As a result, scientists must consider the consequences of their categorizations and rely on ethical values in the process.

This essay draws on philosophers such as Richard Rudner, Heather Douglas, Ian Hacking, Helen Longino, and the WHO’s ICF framework to make two key arguments: (1) judgment is unavoidable when classifying disability, and (2) objectivity in science is actually improved when these judgments are made explicit and open to critical review.

The first reason judgment is unavoidable is that no evidence determines its own meaning or sufficiency. A measurement or observation may provide information, but scientists must still decide what claim that information supports. The same evidence can often support different conclusions depending on the conceptual framework through which it is interpreted. For example, a rise in average test scores might be attributed to upgraded educational quality, changes in testing methods, or demographic shifts. The data alone do not provide an interpretation or indicate which interpretation is correct.

The same issue arises in how we classify disability. A diagnosis in a clinic may show a patient has problems with communication, mobility, or sensory processing. Yet these observations alone do not justify a disability classification. Take vision impairment. Some people have poor vision without being blind. Others, considered legally blind, still have some vision. Two students may both struggle to read the board in the classroom. Yet, one is labeled disabled, and the other simply wears glasses. These cases show that classification involves interpretation, not just

observation. Scientists must decide which traits are relevant, how to measure them, and how meaningful the symptoms are in relation to larger questions about function and participation.

This need for interpretation becomes even more important because disability is not simply a medical condition but an interaction among individuals and their environments. A mobility impairment may create severe restrictions in inaccessible environments, while posing fewer barriers in environments where assistive technologies and accessible infrastructure are available. Likewise, a person may perform adequately in a structured assessment yet experience substantial difficulties in everyday social or instructional settings. Because disability involves multiple dimensions of evidence, no mechanical rule can determine how all relevant factors should be weighed. Judgment is therefore not an optional addition to scientific reasoning; it is part of the process through which observations become evidence capable of supporting a claim.

Furthermore, disability classification involves ethical judgment because it concerns conditions that society may need to accommodate. Richard Rudner's essay "The Scientist Qua Scientist Makes Value Judgments" argues that evidence never completely eliminates uncertainty and that judgment is therefore unavoidable. Rudner explains that scientific hypotheses are rarely established with complete certainty. Scientists must decide when the available evidence is sufficient to accept a hypothesis, and this decision cannot be made by evidence alone because some possibility of error always remains. As Rudner states, "How sure we need to be before we accept a hypothesis will depend on how serious a mistake would be" (2). In other words, judgments about evidential sufficiency depend partly on the consequences of being wrong. Heather Douglas extends this argument by explaining why values inevitably become involved in that process.

Heather Douglas develops this point through her concept of inductive risk, explaining why values become relevant to that decision. As Douglas argues, "non-epistemic values are required in science wherever non-epistemic consequences of error should be considered" (559). Values do not determine what scientists believe, nor do they replace evidence. Rather, they play an indirect role in setting evidential thresholds when different kinds of errors carry different consequences. Although Douglas's argument is often discussed within feminist philosophy of science, her concept of inductive risk is highly applicable to disability classification. A false negative occurs when someone who requires support is not classified as disabled, possibly restricting access to accommodations, services, or legal protections. A false positive, however, may undermine a category's accuracy and reduce its usefulness in recognizing specific support needs. Evidence alone cannot determine which error is more serious in every context. Scientists must judge how these risks should be balanced and justify the standards they adopt.

Importantly, neither Rudner nor Douglas argues that scientists should simply choose whichever classification produces the most desirable outcome. Rather, both emphasize that judgments about evidential sufficiency must be justified. As Rudner notes, the seriousness of a possible mistake affects how much certainty is required before accepting a claim (2). The point is not that values replace evidence, but that scientists must openly explain why a particular level of evidence is appropriate given the risks involved.

Another reason judgment can be helpful is that it allows flexibility when disability classifications need to change. Disability is partly shaped by social conditions and may require revision as medical knowledge develops and social understandings evolve. The historical revision of autism classification provides a useful example. Under the DSM-IV, autistic disorder, Asperger's disorder, childhood disintegrative disorder, and pervasive developmental disorder not otherwise specified were treated as distinct diagnoses. The DSM-5 reorganized these categories under the more extensive diagnosis of Autism Spectrum Disorder. According to the American Psychiatric Association, this change reflected evidence that the previously separate disorders were better understood as "a single condition with different levels of symptom severity" (American Psychiatric Association 2).

This revision did not result from a single clear-cut measurement. Instead, researchers interpreted various studies, assessed whether existing distinctions were meaningful, and considered whether a spectrum best fit the evidence. These actions required judgment about evidence integration and the classification's intended purpose.

The autism case also demonstrates why classification remains historically revisable. Ian Hacking argues that classifications of people are "moving targets" because people respond to the categories applied to them. Once a classification exists, institutions, families, advocacy groups, and classified individuals interact with it, potentially changing its meaning and application. As Hacking writes, "The target has moved. That is the looping effect" (293).

This does not suggest that autism or disability lacks biological reality. Hacking's point is that classifications become part of social contexts, influencing services, perceptions, and identities. These changing circumstances and new evidence require scientists to judge whether current classifications remain effective or need revision. Thus, judgment is essential in evaluating the sufficiency and relevance of evidence when updating classification systems.

A common objection is that if disability classification involves judgment, it risks becoming subjective or politically driven. Fixed diagnostic criteria may seem more objective because they apply the same standards to everyone. Numerical thresholds and uniform standards can indeed promote consistency and reduce arbitrary decision-making. However, fixed rules do not eliminate judgment; they merely conceal it. Someone must decide which criteria belong in a classification, how they should be measured, and where thresholds should be placed. Therefore, I still hold my claim that judgment is inevitable in science. Sometimes these standards or fixed rules are accepted without question simply because they have been used for a long time or are assumed to be proven and correct. Yet these decisions still reflect judgments about relevance, normal functioning, and acceptable levels of risk. A numerical cutoff may apply those judgments consistently, but consistency alone does not guarantee objectivity.

In fact, hidden judgments may be more problematic than explicit ones because they are harder to criticize. When standards appear to emerge directly from evidence, the assumptions underlying them can become invisible. Explicit judgment is preferable because it allows scientists to explain why particular standards were adopted and enables others to evaluate those reasons.

Helen Longino's account of objectivity provides a scheme for understanding why transparency of judgment strengthens rather than weakens scientific objectivity. According to Longino, objectivity depends not on individual scientists eliminating all values but on scientific communities maintaining "venues for criticism, uptake of criticism, public standards, and tempered equality of intellectual authority" (The Fate of Knowledge 135). Scientific claims become more objective when assumptions can be challenged, criticisms are taken seriously, and standards are publicly justified.

Applied to disability classification, Longino's view suggests that scientists should make their judgments visible rather than pretending they do not exist. Researchers should explain what evidence they consider relevant, why they regard it as sufficient, and what risks informed their decisions. As historically shown, people with disabilities and advocacy communities have contributed to this process by identifying overlooked evidence and questioning assumptions embedded in existing classifications. Their perspectives do not automatically determine scientific conclusions, but excluding them would weaken the critical scrutiny that objectivity requires. Because disability is such an interconnected concept, it benefits from criticism from multiple standpoints and an openness to adapt as social conditions change. This openness helps classifications remain both valid and useful. This argument does not imply that all judgments are equally valid or that scientific classifications should simply reflect political preferences. Judgment contributes to objectivity only when it stays constrained by evidence, publicly justified, and open to criticism. Nor does it imply that disability categories should always become broader or more inclusive. The central claim is more modest: because evidence does not determine its own meaning or sufficiency, and because disability classifications integrate multiple forms of evidence that change over time, judgment cannot be removed from the classification process.

Scientists, therefore, cannot avoid judgment when assessing evidential sufficiency. Disability classification makes this especially clear because it entails integrating biomedical, functional, social, and historical evidence while balancing the consequences of classification decisions. Rudner and Douglas show that evidential thresholds inevitably involve judgment under conditions of uncertainty. The WHO's ICF framework illustrates that disability cannot be reduced to a purely biomedical category, while the history of autism classification and Hacking's looping effects illustrate why classifications remain revisable over time. Although judgment may appear to threaten objectivity, Longino's account shows that transparent and accountable judgment is actually a condition of objectivity itself. In disability classification, judgment is unavoidable, yet objectivity is achievable not by eliminating judgment but by making it visible.

Works Cited

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